Practice workbook · Convert between pie-chart sectors, percentages and counts

A pie chart or circular cycle diagram is read, completed or drawn, with sectors as angles, percentages or time fractions.

4 question parts: 1 worked-example retry and 3 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-MATH-PIE-CHART-SECTORS

1. Question 8 · part b.i · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.i only.

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8 (a) Read the passage below.

Use the information in the passage and your own knowledge to answer the questions that follow.

The human body has physical and chemical barriers that reduce the risk of pathogens entering the body. These barriers are known as the first line of defence. If the barriers are penetrated, then the second line of defence comes into action.

The cells involved in the second line of defence form part of the non-specific line of defence. They do not target specific pathogens but they play an important role in stimulating cells involved in the next stages of defence against disease.

The third line of defence involves specific immune responses by cells. These cells rely on the presence and detection of marker chemicals found on the surface of microscopic invaders. The cells of the specific immune system have receptor proteins that interact with the marker chemicals. The marker chemicals cause the cells to become activated and produce antibodies. The production of antibodies in this way is known as natural, active immunity and results in long-term immunity. However, there are other ways that antibodies can provide protection against disease. Natural passive immunity and artificial passive immunity are two similar processes that minimise the risk of too much harm being caused by pathogens in the body.

(i) Name a physical barrier that prevents the entry of pathogens into the body. (lines 1 to 3)

(1)

(ii) Suggest how chemical barriers can reduce the harm caused to the body by pathogens. (lines 1 to 3)

(1)

(iii) Phagocytes are one type of cell involved in the second line of defence.

Describe the process of phagocytosis. (lines 4 to 6)

(2)

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1. Question 8 · part b.i · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.i only.

(iv) State the scientific name for the marker chemicals found on the surface of microscopic invaders. (lines 7 to 11)

(1)

(v) Suggest what is meant by a specific immune response. (line 7)

(2)

(vi) Describe the differences between natural passive immunity and artificial passive immunity. (lines 14 to 16)

You should give examples in your answer.

(2)

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1. Question 8 · part b.i · 3 marks

June 2025 Paper 2R

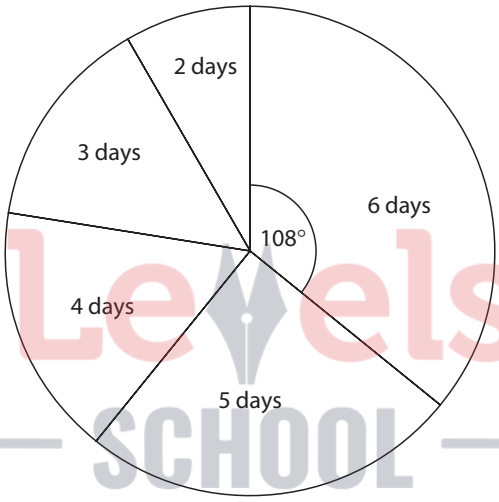
Worked-example retry · Attempt part b.i only.

(b) Scientists test how quickly 30 different types of antibiotic reduce the population size of a resistant strain of bacteria.

The scientists record the number of antibiotics that reduce the population of the resistant strain of bacteria by half.

They do this after 2 days, 3 days, 4 days, 5 days or 6 days.

Each section of the pie chart represents the number of antibiotics.



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(i) Calculate the number of types of antibiotic that reduce the population size of the resistant strain of bacteria by half in 6 days.

(3)

number of types of antibiotic = ...

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1. Question 8 · part b.i · 3 marks

June 2025 Paper 2R

Worked-example retry · Attempt part b.i only.

(ii) State a reason why it is important to wear gloves and a face mask when investigating bacteria.

(1)

(iii) Describe how the use of antibiotics can cause antibiotic-resistant strains of bacteria.

(2)

(Total for Question 8 = 15 marks)

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Working and answer

2. Question 4 · part a.i · 2 marks

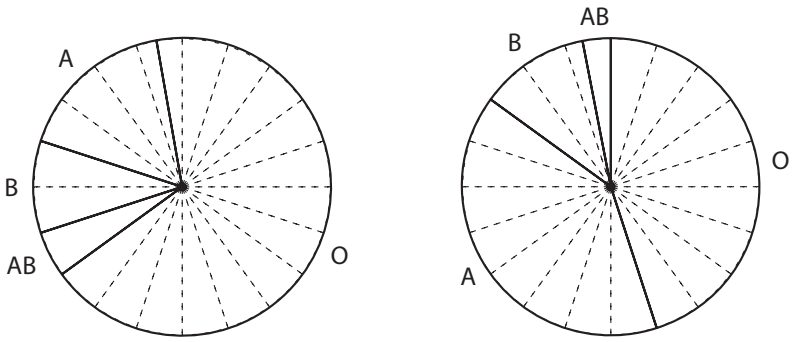
June 2022 Paper 1

Further practice · Attempt part a.i only.

4 (a) Humans belong to one of four blood groups.

These blood groups are A, B, AB and O.

The pie charts show estimates of the percentages of people from Africa and the percentages of people from Europe in each blood group.



(i) Complete the tables to give the percentage of each blood group.

Some information has been given.

(2)

Africa

Blood group	Percentage of blood group
O	68
A	
B	
AB	5

Europe

Blood group	Percentage of blood group
O	45
A	
B	12
AB	

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2. Question 4 · part a.i · 2 marks

June 2022 Paper 1

Further practice · Attempt part a.i only.

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(ii) Some people need a blood transfusion.

Explain why people with blood group A cannot receive a blood transfusion from people with blood group B.

(3)

(iii) A male with the genotype I^AI^O and a female with the genotype I^BI^O have a child with the genotype I^OI^O.

Draw a genetic diagram to show how the child inherits the genotype I^OI^O.

(3)

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(iv) Determine the probability that a child from these parents will inherit the blood group AB.

(1)

probability = ...

2. Question 4 · part a.i · 2 marks · Working and answer

Working and answer

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3. Question 6 · part a.i · 2 marks

November 2020 Paper 1R

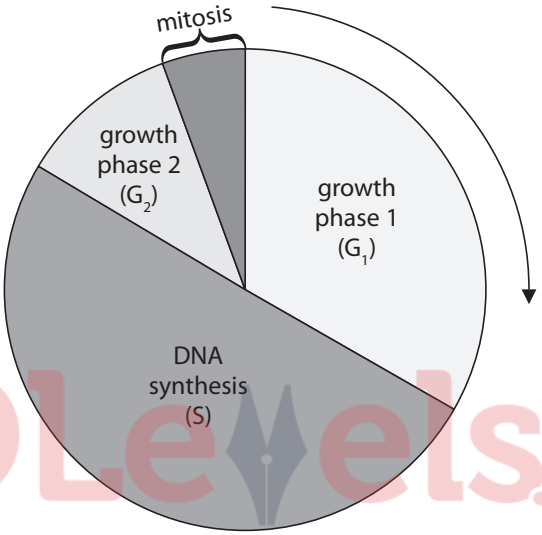
Further practice · Attempt part a.i only.

6 (a) Cells in the body go through several cell cycles in their lifetime.

Each cell cycle is divided into different stages.

Different events take place in the cell during each stage.

The diagram shows one cell cycle of a body cell.



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(i) One cell cycle takes 24 hours to complete.

The total time spent in G_1 and G_2 is ten hours.

Calculate the time spent in mitosis.

(2)

time = ... hours

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3. Question 6 · part a.i · 2 marks

November 2020 Paper 1R

Further practice · Attempt part a.i only.

(ii) Each of the phases of the cell cycle prepare the cell for mitosis.

Suggest the events that take place in the growth phases of the cell cycle.

(2)

(iii) The cell cycle completes by producing two new body cells.

Describe the genetic features of the two cells produced by mitosis.

(2)

(iv) The table lists some of the stages of mitosis.

Complete the table by giving details of the event that takes place in each stage.

(3)

Stage	Events
metaphase	
anaphase	
telophase	

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3. Question 6 · part a.i · 2 marks · Working and answer

Working and answer

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4. Question 4 · part a · 4 marks

June 2023 Paper 1R

Further practice · Attempt part a only.

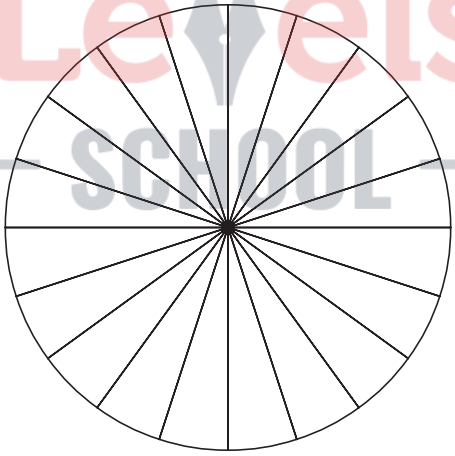
4 The table shows the percentage composition of the diets of two 10 year-old girls.

One girl lives in Area A and the other girl lives in Area B.

Component	Percentage (%) composition of diet	
	Area A	Area B
cereals	75.0	15.0
fruit and vegetables	15.0	37.5
eggs, fish and meat	2.5	27.5
milk, cheese and butter	7.5	20.0

(a) Complete the pie chart to show the diet of the girl in Area B.

(4)



(b) Explain what is meant by a balanced diet.

(2)

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4. Question 4 · part a · 4 marks · Working and answer

Working and answer

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